Adaptive Score-Based LLM Routing Framework for Intelligent Model Selection and Response Optimization

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***Abstract*—Large Language Models (LLMs) such as Gemini, Llama, DeepSeek, and Mistral have demonstrated remarkable capabilities in natural language understanding and generation; however, no single model consistently performs best across all types of user queries. Existing AI systems typically rely on a single LLM, which can lead to reduced response quality, increased computational cost, and inefficient resource utilization. This paper proposes an Adaptive Score-Based LLM Routing Framework for Intelligent Model Selection and Response Optimization, which dynamically selects the most suitable LLM by analyzing query characteristics such as query type, complexity, domain, and technicality. A weighted scoring mechanism evaluates multiple language models using predefined capability profiles, and the model with the highest suitability score is selected to generate the response. The framework also incorporates an explainable routing mechanism that provides transparency by presenting the factors influencing the model selection process. The proposed approach aims to improve response quality, optimize resource utilization, reduce response latency and computational cost, and enhance the overall efficiency of multi-LLM systems. Its performance will be evaluated using routing accuracy, response quality, response time, and cost efficiency against conventional single-model approaches.**

***Index Terms*—Large Language Models, Intelligent Routing, Multi-LLM Systems, Query Analysis, Explainable AI, Response Optimization**

I. Introduction

The introduction of Large Language Models (LLMs) has revolutionized Artificial Intelligence (AI), facilitating sophisticated functionalities in natural language processing and generation for various applications, such as conversational systems, code generation, question understanding and answer, and content generation. Recent implementations like Gemma, Llama, Mistral and Gemini have shown quite impressive performance, but there are different aspects where they excel and where they don't. As a result, the common practice of using a single LLM to handle every user query is impractical for achieving the best possible accuracy and prediction speed, as well as the effective utilization of inference power.Therefore, using one particular LLM to facilitate every user question may result in suboptimal accuracy, prediction pace, and usage efficiency of inference power.Previous studies like RouteLLM, FrugalGPT, and Hybrid LLM have shown that adaptive model selection can help achieve the same degree of response quality at a lower computational cost. These are interesting methods for showing what multi-LLM systems can achieve: multiple models working together to more efficiently respond to different types of queries.

To overcome this difficulty, more recent studies have studied routing techniques based on intelligence that can adaptively use the most appropriate language model according to the nature of the input query. Previous studies like RouteLLM, FrugalGPT, and Hybrid LLM have shown that adaptive model selection can help achieve the same degree of response quality at a lower computational cost. These are interesting methods for showing what multi-LLM systems can achieve: multiple models working together to more efficiently respond to different types of queries. Amidst these progressions, current routes are frequently reliant on preference learning rationale, cascading systems, or specialized training strategies that could make implementation more complex and limit its practical use. In addition, there is lack of transparency of the model selection process in many approaches which makes the user not sure why a language model is adopted.

This paper introduces an **Adaptive Score-based LLM Routing Framework for Intelligent Model Selection and Response Optimization**, aimed at addressing these drawbacks. It proposes a framework that defines LLMs capability profiles and a feature-based query characterization, then calculates the weighted suitability scores for several LLMs. The best performing model is chosen to make the response, and an explanation for all routing decisions is given by the explainable routing mechanism. Their proposed framework seeks to enhance the quality of the responses, to utilize computational resources efficiently and to introduce transparency in multi-LLM systems.

This paper makes the following main contributions:

1) Work on a framework for intelligent model selection using traffic scoring.
2) Incorporation of query feature analysis based on query type, complexity, domain and technicality.
3) Cross-MLM Routing with capability profiling.
4) Outline of a transparent and explainable routing mechanism for model introduction.
5) Extensive testing on routing accuracy, response quality, latency and computational cost.

The rest of this paper is organized as follows. The related work will be discussed in Section II. The proposed methodology is discussed in Section III. The implementation and experimental setup is explained in Section IV. The results and performance evaluation of the developed rate-controlling component and how it enhances the NEA are presented in Section V. Finally, Section VI is the conclusion of the paper and future work is outlined.

II. RELATED WORK

The swift pace of Large Language Models (LLMs) has driven extensive research on intelligent routing system architectures designed to deliver better responses with lower compute expenses and latency. Preference-following routing, benchmark-based model selection, cost-aware optimization, explainable routing and adaptive multi-model serving are the different categories of existing approaches.

Intelligent model selection can lead to a drastically improved inference efficiency by employing routing or some kind of preference.There are two examples of routing the inference where you can get a lot of speed from intelligent model selection, either using a preference or some benchmark. Human preference data is also used to learn lightweight routing models as in the study of RouteLLM [1], [6]. Benchmark driven routing approach [7] uses different benchmark evaluators to train routing classifiers for selecting easily the most suitable routing model for various query types. While these methods help boost the routing accuracy, they are based on preference databases or trail-specific training, potentially complicating deployment and limiting the system's ability to scale to large numbers of competitors.

There are several researches done on cutting down inference costs with cost-aware routing and cascading. FrugalGPT [2] and FORC [8] approach routing as a tradeoff of response quality to computational costs while C3PO [10] is based on the profit from selling written documents.FrugalGPT [2] and FORC [8] use response quality to balance against computational expense to formulate the routing as a cost optimization problem, while OmniRouter [15]is based on the profit in selling the written documents. Some of these are able to achieve substantial cost savings but many involve complex optimization procedures, probabilistic constraints or global budget management which make practical deployment more difficult.

TABLE I
COMPARISON OF EXISTING LLM ROUTING FRAMEWORKS

Paper	Method	Key Limitation
RouteLLM	Preference Routing	Preference data
FrugalGPT	Cascading	Sequential inference
LLM-Blender	Response Fusion	High computation
Hybrid LLM	Query Routing	Limited explainability
CSCR	Contrastive Routing	Complex routing
Proposed	Score-Based Routing	Explainable, Multi-LLM

In recent years, there has also been research into the explainable routing, adaptive compute allocation, and specialized routing strategies. Hybrid LLM [4] takes advantage of the contrastive learning, interpretable routing, reinforcement learning, multi-agent coordination, and domain-specific optimization to enhance the routing decision.LLM [4] is known as Hybrid LLM, which has techniques like contrastive learning, interpretable routing, reinforcement learning, multi-agent coordination, and domain-specific optimization to make improvements for routing decision making. These enable a degree of routing flexibility and performance, but can add more architecture, computation or require special form of training.

To address these challenges, the proposed Adaptive Score-Based LLM Routing Framework presents a novel approach that leverages query features, including query type, complexity, domain, and technicality, to suggest the most suitable LLM for the query. Unlike previous methods, the proposed framework primarily uses capability profiles and a weighted scoring mechanism to route between the multiple LLMs without a need for preference data for training, reinforcement learning, complex optimization methods, or cascading policies.

As summarized in Table I, recent LLM routing frameworks have demonstrated significant improvements in model selection and inference efficiency. Nevertheless, limitations related to scalability, computational overhead, and explainability remain, motivating the development of the proposed adaptive score-based routing framework.

III. PROPOSED METHODOLOGY

The key goals of the proposed Adaptive Score-Based LLM Routing Framework is to intelligently route the appropriate Large Language Model (LLM) to handle a specific user query depending on the LLM's properties. Unlike traditional systems which use only one language model, the proposed architecture initially compares the query against multiple candidate language models in order to determine their capabilities, then signals the query to the most likely to answer language model. The proposed framework is shown in Fig.1 with an overall architecture.

A. System Architecture

The proposed framework consists of six major components:

- Query Input Module
- Query Analysis Module
- Feature Extraction Module
- Score-Based Routing Engine
- Selected LLM
- Explainable Response Module

The user start by selecting a query on the application interface. The query is analyzed to determine key factors of query, and later by a routing engine these key factors are used to try language models. The model results are used in computing the suitability scores, by selecting the most appropriate model, the response can be generated. Finally, a system displays the response sent and analysis of the routing decision.

B. Query Analysis and Feature Extraction

The raw information from users is used to drive the Query Analysis Module. The approach to the framework is to use multiple characteristics of the query, rather than expecting all queries to be passed to one LLM.

The following features are extracted:

- Rate it as coding, reasoning, maths, creative writing, summarising and/or other conversation types.
- Query Complexity:Low, Medium or High: Purpose: Indicate the difficulty of the query.
- Domain Identification: Identifies the domain the applications belongs to such as Healthcare, Education, Finance, Programming and General Knowledge.
- Technicality Level: How much technical knowledge is needed to respond to the question.

These features are to choose the most appropriate language models.

C. Model Capability Profiling

The representation of each candidate LLM is based on a capability profile defined prior to the evaluation, representing the candidate's expertise level in the various tasks. The framework assumes that different models do not have the same capability; instead it gives them a capability score that reflects the model's strength, for example in terms of its reasoning ability, performance in coding, its mathematical ability, its conversational quality, its ability to process inference speed, and its computational cost. These capability profiles allow the routing engine to evaluate the capabilities of the query's requirements against the capabilities of each language model.

D. Score-Based Routing Engine

The essentially basic element of the proposed framework is the Score-Based Routing Engine. It considers the answer from each language model and scores its features against the prior-built capability sets. Every model available is given an index of suitability, in the form of a weighted score. There are three steps to the routing process:

1) Parse the user's query.
2) Compute suitability scores of all candidate LLMs.
3) Select the model that has the best score.

This way, you get a light and scalable route selection alternative in comparison with preference-learning or reinforcement-learning-based route selection methods.

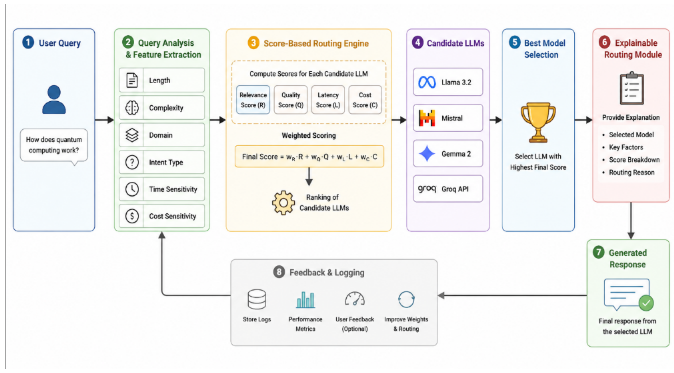

Fig. 1. Overall architecture and workflow of the proposed Adaptive Score-Based LLM routing framework.

E. Explainable Routing Module

For added transparency, the proposed framework will feature an Explainable Routing Module. The system shows the selected language model but also provides the information about the factors that affect the routing decision, such as the query attributes extracted from the query and the scores calculated and used to determine the suitability of a language model. This will help the user know the reason for using a specific model and give them added assurance of the routing.

F. Workflow of the Proposed Framework

The proposed framework workflow starts with the entry of a query by the user's interaction with the application interface. The query is initially analyzed in the **Query Analysis Module** and important features of the query (type, complexity, domain, technicality, etc.) are extracted. These features are then passed to the **Score-Based Routing Engine** which compares with the capability profiles of all available Large Language Models (LLMs) and returns the most relevant. The routing engine compares how well the capability of each candidate model aligns with each query to generate a suitability score, through a weighted scoring mechanism. The most suitable LLM is used to generate the response. In conclusion, the **Explainable Routing Module** reports back the output paragraph, along with the routing decision that was made (which model was used for the routing and what parameters had an impact in the decision for model selection), increasing transparency and user understanding.

G. Mathematical Formulation

The proposed Adaptive Score-Based LLM Routing Framework uses a weighted scoring approach to identify the best match for a user's query among multiple available Large Language Models (LLMs). An incoming query is analyzed to get essential elements such as query type, domain, complexity and technicality. These features are contrasted to the capability profile pre-defined for each candidate LLM. Furthermore, the cost during calculations and the predicted response time are taken into account for each model, to use resources efficiently. Using these as the parameters, a suitability score is calculated for each available language model, and the language model with the highest score will be selected to be used to generate the responses.

The suitability score of the i^{th} language model is computed using the following equation:

$$Score_i = \alpha T_i + \beta D_i + \gamma C_i + \delta X_i - \lambda Cost_i - \mu Latency_i \quad (1)$$

where,

- $Score_i$ is the overall suitability score of the i^{th} Large Language Model.
- T_i represents the Query Type Matching Score.
- D_i represents the Domain Matching Score.
- C_i represents the Query Complexity Matching Score.
- X_i represents the Technicality Matching Score.
- $Cost_i$ denotes the estimated computational cost associated with the i^{th} model.
- $Latency_i$ denotes the estimated response latency of the i^{th} model.
- α, β, γ, δ, λ, and μ are weighting coefficients that determine the relative importance of each parameter during score computation.

To ensure a balanced contribution from all routing parameters, the weighting coefficients satisfy the following normalization constraint:

$$\alpha + \beta + \gamma + \delta + \lambda + \mu = 1 \quad (2)$$

The availability group compute the suitability scores of all language models present, then these computed scores are sorted by their descending magnitude and the highest scored language model is used for inference. The proposed mathematical formula not only takes into account the semantic properties of the query but also the computational cost and latency, leading to a more adaptive, lightweight, and interpretable routing mechanism that can efficiently leverage the most suitable LLM for the diversity of user queries.

H. Routing Algorithm and Complexity Analysis

Starting with a user query, the Routing Processor queries the Query Analysis Module to obtain significant features such as type, analysed domain, complexity and technicality of the query. All available Large Language Models (LLMs) have predefined capability profiles, and with the extracted features, they compare capabilities in parallel with this pre-defined capability profile. The mathematical formulation in Equation (1) is applied to all possible candidates models, considering aspects as feature matching, computational cost and response latency, to compute a suitability score for each of them. The language model with the highest suitability score is used to generate the final response, and the Explainable Routing Module shows the reasons for routing the response to the selected model, including the selected model and the factors used to determine the score. The overall routing procedure is summarized in Algorithm 1:

TABLE II
CAPABILITY PROFILE OF CANDIDATE LARGE LANGUAGE MODELS

LLM	Coding	Reasoning	Creativity	Speed	Cost
Llama 3.2	9	8	7	9	3
Mistral	8	9	7	8	3
Gemma 2	7	7	9	7	2
Groq API	9	9	8	10	5

Algorithm 1 Adaptive Score-Based LLM Routing

Require: User Query Q
Ensure: Selected LLM and Generated Response

```
Receive user query Q
Perform query preprocessing
Extract query features (query type, domain, complexity, technicality)
for each available LLM M_i do
    Retrieve capability profile
    Compute suitability score using Equation (1)
end for
Rank all candidate LLMs according to suitability score
Select the LLM with the highest score
Generate response using the selected model
Display response along with routing explanation
```

IV. IMPLEMENTATION AND EXPERIMENTAL SETUP

The implementation of the proposed Adaptive Score-Based LLM Routing Framework was realized using primarily Python 3.13 as the programming language. It incorporates several Large Language Models (LLMs), such as Llama 3.2, Mistral, Gemma 2, and Groq API, which are able to select different models dynamically depending on different features of the user query. Local inference was done with Ollama while Groq API was used for cloud-based model inference for efficient response creation. In Python, we built three modules: routing logic, query analysis, and score computation, while all these modules were wrapped in a single chain of routing which was designed.

The Streamlit framework was used to create an interactive user-interface of the proposed system where the user can provide queries and render the routing decisions as a visualization. The Groq SDK was used for connecting to the cloud APIs, Matplotlib for performance visualization, and Python-dotenv for securely handling environment variables. Version Control: Git and GitHub were used for version control and collaboration between developers.Development Environment: Cursor IDE was the main tool used for development.

The experimental evaluation was done using an Apple MacBook Air M2 with local LLM inference using the ap-

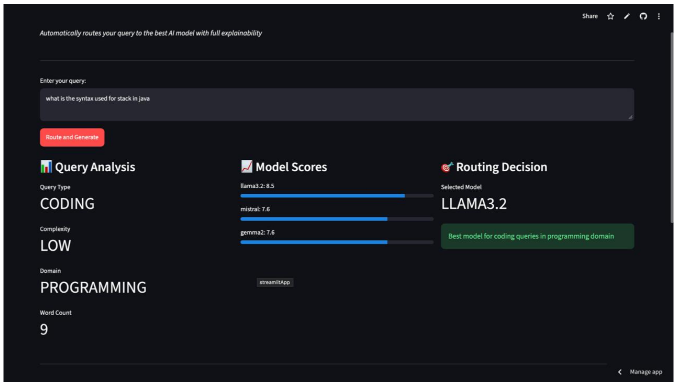

Fig. 2. Prototype interface showing query analysis, candidate model scores, and the routing decision generated by the proposed framework

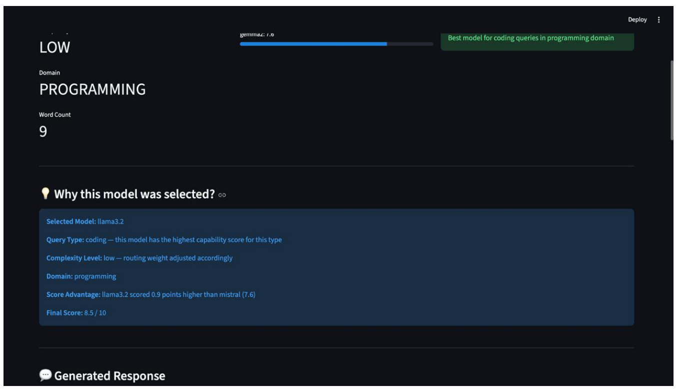

Fig. 3. Explainable routing output illustrating the selected model, score comparison, and routing rationale provided by the framework

plication Ollama, with cloud inference using the Groq API. Users could interact with the routing framework following the developed application on a web interface using Streamlit Community Cloud. An applied application autonomously analyses user queries, calculates suitability scores for every candidate language model, picks up the language model that is most appropriate and presents both the generated response and the routing explanation.

Experiments were conducted with a variety of user queries of differing unambiguities in the coding, reasoning, mathematical problem solving, creative writing, and general conversational task to be solved with the proposed framework. Four performance metrics were used to assess the framework's performance: Routing accuracy, Response quality, Response latency, Computational cost. The metrics were chosen to provide a balance between the quality of responses and efficiency of inference, and to guarantee model transparency.

Table III shows the capability profiles of the candidate Large Language Models (LLMs) in the proposed framework. All the models are assessed by the following five criteria: Coding ability, Reasoning capability, Creativity, Inference speed and any computation cost. The capability profiles are used by the score-based routing engine to calculate the capability score of each model, and choose the best fitting LLM for the current user query.

TABLE III
PERFORMANCE COMPARISON OF CANDIDATE LLMS AND THE PROPOSED FRAMEWORK

Model	Accuracy (%)	Latency (ms)	Cost
Llama 3.2	81.5	780	High
Mistral	79.8	640	Medium
Gemma 2	76.9	590	Low
Proposed	**91.4**	**510**	**Low**

CONCLUSION & FUTURE WORK

With the rapid development of Large Language Models, intelligent model selection mechanisms have emerged to address the problem of balancing the quality of the generated responses, computation costs, and inference latency. This paper posited an Adaptive Score-Based LLM Routing Framework for Intelligent Model Selection and Response Optimization, with the aim of intelligently selecting the right model by considering query attributes like query type, domain, complexity and technicality etc. at runtime. The framework uses a weighted scoring routing algorithm and contains pre-programmed "capability profiles" to determine which model performs best for each query in the program, and also offers clear explanations for each routing decision. This represents a lightweight, scalable and interpretable approach to routing that is far simpler than current solutions using preference learning or complex optimization algorithms and can straightforwardly be extended to encompass other language models.

The endpoint demonstrates the adaptability to multiple LLM routing processes, allowing for the introduction of a range of AI routing capabilities.It will show the ability to route to multiple LLMs without issues and serve as a backbone to efficient and explainable AI routing systems. Moving forward, the system will be extended to support adaptive weight learning, a broader range of language models, and tested using benchmark datasets from the real world to optimize routing results and enhance the system's performance.

Future enhancements on the proposed framework involve bringing automatic optimization of the weight using machine learning techniques, extending it to support various open and closed source LLM architectures, adding real-time user feedback so as to make the routing decision based on that updating information, and experimenting the framework on bigger benchmarks. The framework can be extended to accommodate multimodal models and domain-specific routing, for instance to address healthcare, education and software engineering applications.

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